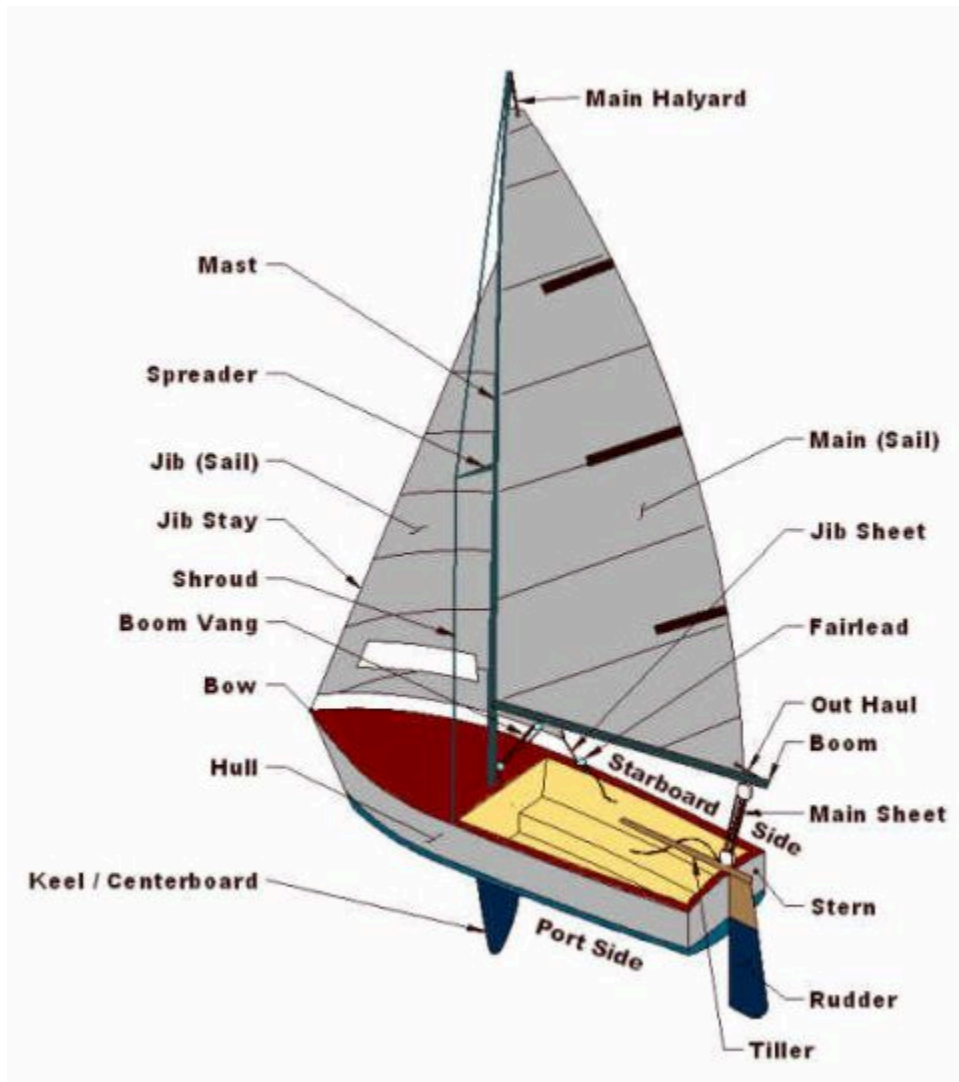


HOW TO SAIL 101



This image is crucial and is referenced a few times.

Warning/Introduction

Welcome to intro to sailing! This is an overarching guide for someone who has seen sailing before/and or is vaguely familiar with boating and/or being on the water. This being said, anyone with determination can learn from zero to having a basic understanding of sailing with this guide. But, before doing anything on the water there are a couple things that you must be aware of to sail. First, if you're on the water you should be able to swim, at least at a basic level. A good

metric is you're able to tread water for a little while (5 minutes). PFDs are a must almost always but you should still be able to swim. Furthermore, sailing is a fast and possibly harmful activity, especially depending on the size of waves, speed of the wind, rate of changing conditions, etc. Sailing can be done in a variety of weather, but there are better days than others for beginners, and you should be aware of this fact. That being said, done in safe conditions and with confidence in your understanding, sailing is a relaxing sport that most people are capable of doing well.

Step 1: PFD

The first step in sailing the smallboat sailboat is actually way before you actually get into a boat. There are a few steps that need to be addressed that are preliminary, mostly safety and maintenance related. We must address the lifejacket/floatation component first. This is mostly for your ability to be comfortable within the water if something happens while out on the water (capsizing, a crew member falls out, you fall out, centerboard or rudder issue within the water, etc). As previously mentioned, you should be able to swim, but the PFD is important because in the case you capsize (we will touch on this later).

Step 2: Rigging/Sail Check

The next step is taking a look at the sails, lines (ropes designed to hold the sail), blocks (pulleys that allow you to tighten the lines with more ease) and other rigging on the boat itself. Is anything broken, frayed, or severed? If not, you're clear to start the rigging! Each boat is different, but the manual or someone who knows the sailboat would be able to assist you in this

process. Usually it involves raising the jib (front sail) and main sail using the halyards, which are lines that use the blocks we mentioned earlier to raise the sails and take all slack out of the line and make sure the sails are taught. With your sails raised and your lines organized and checked for safety, you're off to get on the water!

Step 3: First Sailing Steps

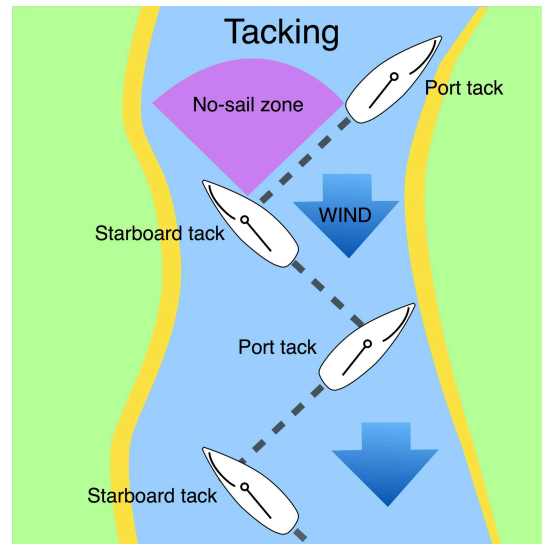
The first step is put on the rudder/tiller system (depends on the boat but is straightforward). Next is when you're in deep enough water, lower your centerboard. This can be done either with loosening a line or just by pushing it down into the water. This works as a counterforce and allows your boat to not be taken away by the wind laterally and move forward when the wind is not right behind you. Now, grab the tiller and start to steer! The tricky thing is that if you want to go right, you just need to push the tiller left, the input to the tiller is INVERSE to the direction you want to go. It takes a minute or two and then it'll feel natural.



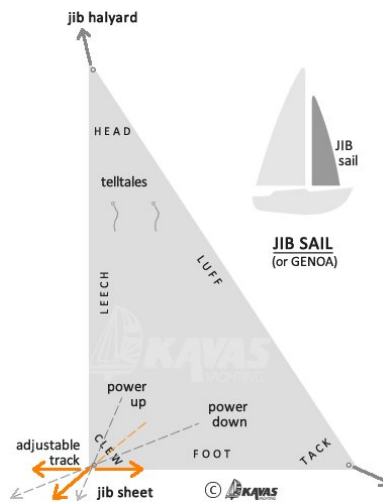
Step 4: Proper Sailing!

Now, tighten the main sheet (line attached to the boom), this will start to utilize the power of the wind to propel you forwards. As the speed increases and the sails fill with air, the wind will start to capsize the boat, to counteract this, just use your weight and lean out to balance the boat out. For an example of this, see above. **Import concept: If you're moving too quickly or your boat is going to capsize due to how much wind you have, let the mainsheet out and lose your speed.** In sailing, especially for fun, getting soaked and having to un-capsize your boat is not worth going insanely fast, if it's not sustainable, let out on the mainsheet and you'll lose your speed and the boat will go back to sitting flat on the water.

Step 5: Tacking/Jibing - Changing Direction

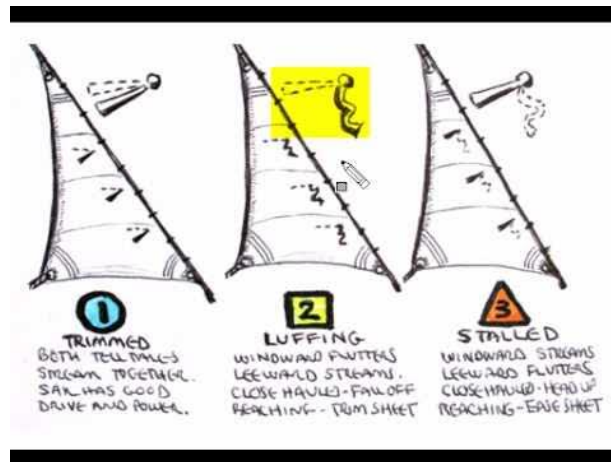


Now, this is almost the most crucial key to understanding how to go anywhere, and get back to where you came from. This is the concept of tacking, which is effectively changing direction to continue moving upwind. As seen in the reference image, there is a no sail zone, which refers to the zone where the wind is coming to the boat too directly to be utilized to move forward upwind. Thus, the idea is to flip back and forth at the right angle to move in the correct general direction upwind. Tacking is usually turning 45+ degrees to move on the same angle against the wind as you were before in the other direction (or on the other tack). Now, when you turn into the wind, you must do so with speed, and when this happens the boom (see above) will swing very quickly onto the opposite side to accommodate the wind. Now, the boom is said to be called this because if you're not careful when you tack and you aren't careful, BOOM and it will hit you in the head, obviously problematic. But, diligence and communication across crew and skipper will very easily prevent this from happening. Now that we've covered tacking (usually the most technical movement) you can sail basically any direction.



The jib is not necessarily the less important sail of the two, and can give you a lot of speed. It does however introduce a lot of complexity into the sailing experience. So much so, that in college sailing usually you have a second person sailing with you to man that sail (among other jobs). Nevertheless, the jib works with two sheets in a give and take style. So when the wind is coming from the port side, you want to loosen that side and tighten the starboard side, so you can capture the wind. Furthermore, when you tack, you just swap sides for the jib. The Jib usually captures air differently, so the jib and the main will be at different visual angles to utilize the same wind from the same direction.

Sidenote: Tell Tales



On any sail, if you want to capture the most wind in either of your sails, look for the little ribbons that indicate where you can gain the most wind in your sails (this is proportional to speed of course) on any given tack.

Step 7: Luffing/Irons/Parking

Now, you've sailed, hopefully got some practice on tacking, using your jib, etc. So, how do you get back to the dock/get off the boat/stop moving. This is a pretty simple idea, you just want to point the bow of the boat directly into the wind. There, no wind can be in your sails and you can stop dead in your tracks. Where is this helpful? Maybe if there is an issue on the boat or you need to pause at a start line in a race, or you want to get off the boat, this is a useful skill.

The next step is getting out of being "in irons" (not moving). This can be done by quickly steering into one direction and following that tack, and then adjusting direction to go where you need to go.